SPE-229733-MS

Alternative Methods to Subsea Pipeline Hydrotesting

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This paper was prepared for presentation at the ADIPEC held in Abu Dhabi, UAE, 3 – 6 November 2025.

Abstract

Hydrotesting has long been a standard pre-commissioning requirement for subsea pipelines, ensuring structural integrity. The entire process of subsea hydrotesting the pipeline requires substantial resources and has a direct impact on the environment due to the disposal of a large volume of inhibited water, which can harm marine life. While hydrotesting has served as a fundamental verification technique, its core limitations include its inability to detect or address non-pressure- related threats, such as axial stresses, external mechanical impacts, and fatigue cracking. Hydrotesting has rarely been identified as a failure mode in modern pipeline laying, largely due to effective offshore welding procedures and controls. However, hydrotesting may also bring a false sense of security, as crack growth is time dependent.

With advancements in materials science, Non-Destructive Testing (NDT), and digital analytics, a viable alternative to pipeline hydrotesting has been developed for large-diameter pipelines in recent years. This necessitates a re-evaluation of traditional practices within the Middle East oil and gas sector as the industry pursues more efficient and environmentally conscious execution.

This paper presents a comprehensive review of the technical, economic, and environmental literature on alternative methods for hydrotesting, supported by industry standards (DNV-ST-F101, DNV-RP-F115), case studies, and risk- based assessments. Key findings highlight that mill hydrotesting, in-line inspection (ILI), predictive data-driven analytics, and quantitative risk analysis (QRA) with ALARP (As Low As Reasonably Practicable) collectively achieve superior safety levels compared to traditional hydrotesting. This study also quantifies benefits, including CAPEX reduction, schedule savings in offshore operations, and, more importantly, significant CO2 reduction per test, reinforcing the industry's shift toward People, Planet, and Profit (PPP) objectives.

Guidance from the REPLACE (a DNV Joint Industry Project) complements Industry Standard DNV-ST-F101 by providing practical insights into implementing alternative integrity verification methods for pipeline commissioning. By embracing these innovative strategies, Pipeline EPC Contractors can achieve enhanced operational efficiency while also realizing measurable environmental benefits, thereby contributing directly to the achievement of the Sustainable Development Goals

Keywords: Subsea Pipeline, Hydrotesting, DNV-ST-F101, Failure Modes, Risk Analysis, QRA, Sustainability, Subsea Pipelines

Introduction

Hydrotesting has long been a standard procedure for validating the structural integrity of subsea pipelines since the 1950s. This process typically involves first flooding the pipeline with chemically inhibited water, pressurizing the line, holding for a stipulated period, normally 8-24 hours, followed by depressurization and finally flushing the chemically inhibited water into the open Sea.

The scope of this paper is to review the literature and industry practices surrounding subsea pipeline hydrotesting, identify common test failures, and explore inherent technologies that could mitigate the need for flooding and wet hydrotesting, hydrotesting specially from EPC Contractor's perspective while ensuring the detection of bursting strength / leak of pipeline is not compromised.

This paper focuses on assessing the PPP (People, Planet, and Profit) framework, with special attention to the Environmental Aspect. It further explores the DNV-ST-F101 guidelines for waiving hydrotests and highlights the role of alternative procedures such as Hazard Identification (HAZID) and Failure Mode and Effects Analysis (FMEA).

Literature Review

Traditional Hydrotesting: History and Methodology

Current guidelines for system pressure test values are provided in the DNV-ST-F101 and ASME Standards (based on the Operator, which normally ranges from the Design Pressure to 1.5 times the design pressure, not exceeding the hoop stress equivalent to 90% of the SMYS). With offshore pipelines laid with improved welding technology offshore and good control on Weld defects, extremely low failure rates have been reported over the last 20 years (Cosham et al. (2006)).

Despite its widespread use, field hydrotesting has limited value in ensuring the overall integrity of a welded Pipeline system offshore, while it introduces several complications during the EPC phase. The logistical complexity of mobilizing vessels, pumps, and personnel in remote offshore environments contributes significantly to project overheads. The method also raises environmental concerns, particularly with water disposal, chemical contamination, and the potential for accidents during pressure testing. The discharge of large volumes of inhibited water can disrupt marine ecosystems, leading to regulatory scrutiny and increased public concerns.

Over the year, the failure of the Pipeline during hydrotest has been reduced. The majority of the failures reported were due to reasons that could not be identified during the field hydrotest. According to PARLOC, data on the failure rates of the Pipeline have been reported (PARLOC, 2001).

The primary failure types and contributing factors for subsea pipelines are mostly

- Corrosion: (27-40%) Internal, external, or microbiologically induced (MIC).
- Weld Imperfections: (~33%) e.g., lack of fusion, cracks.
- Material Defects: (~14%) e.g., inclusions, laminations.
- External Interference: (10-15%) Incidents from anchor drops or fishing gear.
- Operational Errors: (5-10%) Resulting from over-pressurization or poor maintenance.

Alternative Testing Methods

In recent years, technological advancements have paved the way for alternatives to traditional hydrotesting. This potential paradigm shift leverages sophisticated digital and sensing / detecting capabilities to offer more precise and less intrusive integrity verification techniques. Inline Inspection (ILI) technologies, such as smart pigs and ultrasonic testing, offer the potential to perform pipeline integrity assessments without the need for flooding. These technologies can detect issues such as corrosion, cracks, and material defects in real-time. The high-resolution data gathered by these tools allows for a comprehensive understanding of the pipeline's condition, enabling proactive maintenance and extending asset life. Several studies have highlighted the advantages of using ILI technologies, including reduced operational costs, shorter testing times, and minimal environmental impact (Pieter Swart, 2020; Jones et al., 2023). Furthermore, the ability to integrate this data with predictive analytics / digital twin systems promises a continuous integrity management approach, moving beyond conventional hydrotesting.

The proposed application of advanced technology is considered effective in ensuring the integrity of the Pipeline System without requiring a field hydrotest (Ref. DNV-ST-F-F-101, F-115, DNV JIP Replace).

- Waiver Approach: ILI (smart pigs), 100% NDT (AUT, RT), ECA, AI analytics, IVB review. Meets DNV- ST-F101, but chemical disposal is scrutinized (DNV-RP-F115). Equivalent to hydrotesting, with failure probability <10-5 per km-year via QRA.
- Controls: 100% NDT (AUT, RT), ECA, ILI ensure weld integrity (no cracks, <25 mm lack of fusion).
- Stakeholder Perception: Growing acceptance (e.g., Nord Stream 2).
- Regulatory Compliance: DNV-ST-F101 allows waivers with IVB-verified dossier (NDT logs, ECA rep.

Figure-1, presents expected reduction of risk with waiver of pipeline hydrotest with pertinent advance ILI and NDT application as per DNV standards

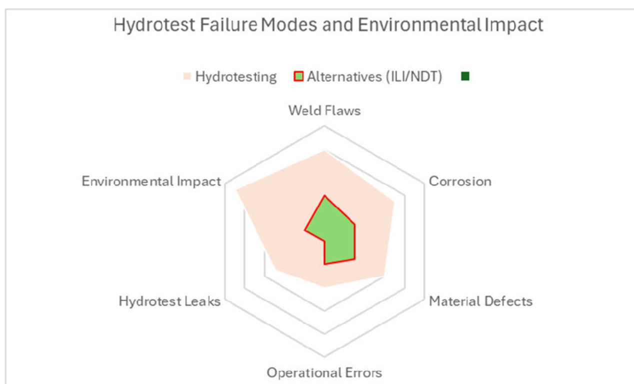

Figure 1—Hydrotest Failure Modes and Environment Impacts.

A detailed assessment for the Alternative method to adopt a waiver of the Pipeline Hydrotest has been presented in Table 1, below:

Table 1—Alternative Procedure to adopt waiver of Pipeline Hydrotest

Parameters	Requirement Description	Compliance Evidence
Threat & Failure Mode Assessment	Systematic assessment of failure modes mitigations for all credible threats.	FMECA Report, Risk Register, HAZID/HAZOP logs.
Reliability-Based Assessment	Lower probability of failure with DNV-ST- F101 safety classes (e.g., 10^{-5} per km- year).	Reliability Report / Risk Acceptance Criteria.
Material Manufacture	100% NDT of longitudinal welds. Enhanced traceability, heat numbers, certificates.	Mill Test Reports, NDT Logs. Mill Certificates, Manufacturing Dossier
Fabrication & Welding	Qualified welders, WPS/PQR in place. 100% AUT/RT + MPI/PT on girth welds. Repair and rework statistics within acceptable limits.	Welding Qualification Records. Weld Logbook, NDT Reports Welding Repair Summary.
Installation Control	Pipe handling and stress control monitoring. Control of tie-in welds (100% NDT, PWHT if applicable).	Bending Logs, Reel Lay Tension Records. Field Weld Reports.
Fracture Mechanics and ECA	Flaw acceptance criteria based on ECA Fracture toughness and CTOD data used. Fatigue and flaw growth analysis over design life.	ECA Report, BS 7910 / DNV-RP-F108 Analysis. Material Test Reports Fatigue Life Assessment.
Documentation & Independent Verification	Third-party IVB review and approval. Full dossier with material traceability, weld logs, ECA, NDT, design.	IVB Compliance Statement, Review Reports. Waiver Dossier Index.
Risk Management	Formal risk assessment with comparison to hydrotest. ALARP demonstration for risk levels. Contingency / Emergency plans in case of leak.	QRA Report or Risk Comparison Table. ALARP Justification, Cost-benefit Study (if applicable). Emergency Response Plan.

The Role of Advanced Risk Assessment Techniques

Risk assessment methodologies, including Quantitative Risk Analysis (QRA) and Strengths, Weaknesses, Opportunities, and Threats (SWOT) matrices, are commonly used to evaluate the effectiveness and safety of alternative testing methods.

These techniques help identify the potential risks associated with replacing traditional hydrotesting and provide a framework for comparing the benefits and drawbacks of various alternatives. Previous studies have demonstrated that combining ILI technologies with detailed risk assessments can provide a comparable level of safety while reducing the overall cost and time required for testing (Smith et al., 2021). This robust analytical framework is crucial for demonstrating that alternative approaches maintain, or even enhance, the integrity and safety standards traditionally assured by hydrotesting, thereby justifying the consideration of waivers.

Regulatory Considerations

The DNV-ST-F101 design code provides specific provisions for waiving hydrotests if equivalent safety levels are demonstrated through alternative methods. This code outlines procedures for alternative testing, including HAZID and Failure Mode and Effects Analysis (FMEA), which assess the likelihood and consequences of potential failures. These guidelines represent a progressive shift by regulatory bodies, acknowledging that robust engineering assessments and advanced technologies can offer comparable assurance to traditional physical tests. The code also emphasizes the importance of maintaining a high level of safety and operational integrity throughout the pipeline's lifecycle. Several industry projects, such as the Turk Stream and Nord Stream 2, have successfully implemented alternative testing methods (Pieter Swart, 2020). These successful implementations provide good precedents, validating the feasibility and safety performance of alternative methods to hydrotest.

Methodology

Hydrotest waivers are a strategic enabler for sustainable, cost-effective pipeline projects. This study employs a comprehensive methodology to assess the feasibility and justification of hydrotest waivers as a strategic enabler for sustainable and cost-effective pipeline projects. The approach integrates established industry standards with advanced analytical techniques and real-world project insights to build a robust framework for decision-making. The following basis highlights the need for justification of the title 'Alternative Methods to Subsea Pipeline Hydrotesting.

- DNV-ST-F101 and JIP-REPLACE (open-source data) provide robust frameworks for alternatives.
- Combining ILI, NDT, and AI ensures integrity without the need for hydrotesting.
- Risk-based approaches (QRA, ALARP).
- PPP (People, Planet, Profit) benefits are significant and measurable.

Design Code DNV-ST-F101: Alternative Procedures for Hydrotest Waiver

The DNV-ST-F101 design code provides a framework and allows for the use of alternative methods to demonstrate the safety and integrity of the pipeline, provided they meet or exceed the safety standards of traditional hydrotesting. Hydrotest is being replaced by introducing quality procedures / stringent material selection criteria along with additional FMEA requirements, i.e., proactive risk mitigation from design through construction, rather than relying solely on a single, post-installation test. Since all these measures have been integrated into EPC Contractors' normal practice for subsea pipeline construction, the need for hydrotest is under re-evaluation. Figure 2 below presents guidance for waiver of pipeline hydrotest.

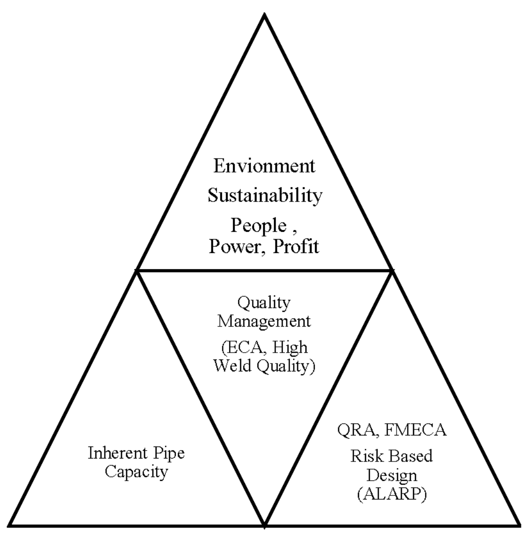

Figure 2—Waiver to Hydrotest Methodology, Ref. DNV-JIP Replace.

SWOT Matrix and QRA for Risk Assessment

A Strengths, Weaknesses, Opportunities, and Threats (SWOT) analysis has been conducted to assess the potential strengths, weaknesses, opportunities, and threats associated with adopting alternative methods to hydrotesting. An attempt to quantify the risks associated with the conventional and alternative methods is provided in Table 2. The risks are marked against impact versus likelihood and increasing impact. The risks associated with the alternate ILI method is considered lower (MEDIUM risk) than the conventional hydrotesting methods (HIGH risk). This quantitative evidence serves as the justification for the adoption of waivers, demonstrating a net reduction in overall project risk. Figure 3 below summarizes the key findings of the SWOT assessment, while Table 2 presents the details of the SWOT analysis.

Table 2—SWOT Analysis for Alternative Methods to System Pressure Test

Strengths	Weaknesses
Reduced CAPEX Cost- CAPEX avoidance by eliminating water procurement, disposal, and downtime.	Initial technology costs (ILI tools) - CAPEX.
Sustainability Gains CO2 reduction: ~500 tonnes per test (avoiding diesel-powered pumps). Water conservation: 12-20 million gallons saved per project. Minimal environmental impact (no water disposal).	Limited effectiveness in certain environments (e.g., deepwater).
Faster testing process, reducing schedule delays, Faster commissioning (30-60 days saved per project).	Technology reliability concerns in some cases.
Real-time pipeline inspection via ILI tools.	Dependence on skilled personnel for ILI operation.
Mill/fabrication testing exceeds hydrotest pressure; digitalization enables predictive integrity.	
Opportunities	**Threats**
Regulatory support for hydrotest waivers (e.g., DNV-ST-F101).	Resistance to change from stakeholders (especially in traditional industries).
Increasing industry adoption of ILI as technology improves.	Potential for technology obsolescence if not upgraded.
Potential cost / time savings on large projects.	Hybrid systems (combining ILI with traditional tests) could incur higher costs.
Hydrotest only verifies hoop stress, not axial loads (e.g., ground movement, corrosion).	Offshore QC gaps may undermine alternative methods.
Water savings (12-20M gal), CO_2 reduction (~500 tonnes), 30-60- day schedule acceleration.	

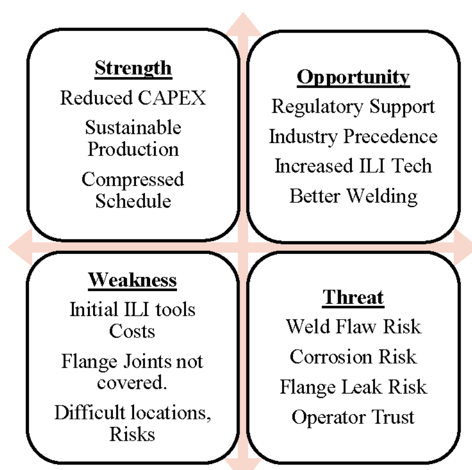

Figure 3—SWOT Chart Summary for Hydrotest Waiver.

Parametric Case Studies to assess induced strain due to Hydrotest

To illustrate the feasibility of alternative methods, the paper includes parametric case studies that examine different scenarios in which hydrotesting is replaced by ILI technologies. By varying key parameters such as pipeline diameter, length, water depth, and material specifications, these studies offer a comprehensive analysis across a range of typical subsea pipeline projects. These case studies assess the risk and reward for EPC contractors, taking into account factors such as pipeline material, location, and project complexity. The results from these case studies provide insight into the potential cost savings, time reductions, and environmental benefits associated with alternative testing methods. Table 4 summarizes the parametric assessment for the induced strains with respect to various pressures for different pipes (diameter / wall thickness) is presented for comparison.

Table 3—Risk Matrix - Hydrotesting vs ILI

Risk Impact \ Likelihood	Low	Medium	High
Environmental Impact	ILI (Minimal impact).		Hydrotesting (High water usage, chemical risk).
Cost		ILI (Moderate initial cost, lower operational cost).	Hydrotesting (High mobilization, manpower costs).
Testing Time		ILI (Faster, real-time inspection).	Hydrotesting (Time-consuming process).
Technology Reliability		ILI (Moderate reliability in remote areas).	Hydrotesting (Proven, but more failures in complex offshore environments).

Table 4—Parametric Pressure Test and corresponding Stain

OD		WT	SMYS	Pressure equal to 100% SMYS Hoop stress	Strain (%)	Hydro Pressure ~90% SMYS	Strain (%)	2*Hydro Pressure	Strain (%)
(in)	(mm)	(mm)	(MPa)	(MPa)		(MPa)		(MPa)	
12	323.9	11.13	450	30.90	0.226	27.83	0.204	55.67	0.408
12	323.9	12.7	450	35.30	0.227	31.76	0.204	63.52	0.409
16	406.4	14.7	450	32.60	0.227	29.30	0.204	58.60	0.408
16	406.4	10.31	450	22.80	0.226	20.55	0.203	41.10	0.407
30	762	19.05	450	22.50	0.226	20.25	0.203	40.50	0.408

A detailed finite element model confirms that the pipe can sustain pressures significantly higher than hydrotest, even twice the hydrotest pressure, and still the induced strain will be in the order of an acceptable 0.4% for workmanship tensile strain limitation. Table 4 presents a parametric pressure test and the correlation between induced strain.

Quantitative Risk Analysis (QRA)

A preliminary QRA was carried out based on indicative likelihood and consequences to verify ALARP Risk and presented in Table 5 below.

Table 5—Quantitative Risk assessment for Initial Riks and Residual Risks

Hazard	Cause	Initial Risk (LxC)	Control Measures	Residual Risk (LxC)	Acceptability
Leak during operation	Undetected flaw in weld.	3 × 4 = 12 (High)	100% AUT, ECA, Fatigue Analysis, IVB Review.	2 × 3 = 6 (Moderate)	Acceptable with ALARP justification
Fracture initiation	Manufacturing defect.	4 × 4 = 16 (Extreme)	Mill QA, 100% NDT, CTOD Tests, Fracture Toughness, ECA.	2 × 3 = 6 (Moderate)	Acceptable
Collapse during installation	Overbend or buckling.	3 × 3 = 9 (High)	Installation monitoring, pipeline integrity checks, real-time stress logs.	1 × 3 = 3 (Low)	Acceptable
Hydrogen-induced cracking (if hydrotested)	Post-hydrotest sour service exposure.	3 × 4 = 12 (High)	Waiving hydrotest avoids HIC risk; + hardness control.	1 × 2 = 2 (Low)	Acceptable
Missed flaw at tie-in weld	Inadequate inspection.	3 × 4 = 12 (High)	Field weld 100% AUT + MPI/PT + IVB witness.	1 × 3 = 3 (Low)	Acceptable

Links to UN Sustainable Development Goals (SDGs)

The current study links to the UN SDGs with a direct impact to the below listed SDGs

SDG 12: Responsible Consumption and Production: SDG 12 has the most direct link to the current study, which discusses "reducing costs and schedules" and "CAPEX reduction,". This speaks to more efficient and responsible use of resources (reducing material and energy consumption associated with traditional hydrotesting). The disposal of used water/effluent is an unsustainable production activity. By proposing alternatives, the paper directly contributes to the environmentally sound management of chemicals and all waste throughout their life cycle, promoting more sustainable consumption and production patterns.

SDG 13: Climate Action: The idea of reducing CO2 per test directly links the work to the Climate Action SDG. This is a clear and measurable contribution to combating climate change and its impacts. Reducing emissions from industrial processes is a key aspect of achieving SDG 13.

SDG 14: Life Below Water: There is an impact on the Environment due to the disposal of a large volume of inhibited water into Marine life. The proposed alternatives aim directly to mitigate this impact, contributing to the conservation and sustainable use of the oceans, seas, and marine resources. This specifically aligns with protecting marine ecosystems from pollution.

SDG 9: Industry, Innovation, and Infrastructure: Instead of relying on the traditional hydrotesting method, our paper proposes a more advanced approach. This shift represents new, innovative ways to build and manage our pipelines, making them more resilient, environmentally friendly, and efficient than ever before.

Results and Discussion

Risk and Reward Assessment

The results from the SWOT analysis (Table 2), the Risk Matrix (Table 3), and the QRA (Table 5) indicate that advanced welding technology and smart Inline Inspection Tools (ILI) present a favorable trade-off in terms of cost, risk, and schedule. In scenarios where the pipeline material and design are optimally suited for effective inspection through ILI, there are potential savings in cost, often reaching millions of dollars, as well as reductions in environmental risks, particularly related to the discharge of contaminated water, and the requirement for additional pumping operations offshore. Furthermore, the time required to complete the integrity verification process is considerably reduced by several weeks or even months for extensive subsea systems, which directly leads to faster project completion and earlier pipeline commissioning, thereby accelerating revenue generation.

Additionally, a hydrotest is not considered a suitable tool for detecting the capacity of a pipe / weld strength. Pipe has an inherent larger capacity to sustain compared to the allowed / normally applicable hydrotest pressure in terms of bursting or plastic deformation. Table 4 presents that even with a hydrotest pressure up to 2 times the design pressure, the induced strain remains in the range of 0.4%, which qualifies not only the elastic behavior of the pipeline but also the workmanship for weld defects.

The primary failure types and contributing factors were weld imperfections, corrosion, and material defects; none of these can be positively detected by field hydrotest, making the field hydrotest an inadequate check.

However, the analysis also reveals that a detailed approach is critical. In certain complex situations, such as when pipelines are in extremely challenging offshore environments with highly irregular profiles, or when the pipeline material presents inherent limitations for current ILI inspection tool capabilities. In such specific cases, a hybrid approach, strategically combining the precision and non-invasiveness of ILI with targeted, localized hydrotesting for specific segments or critical components, could indeed provide an optimal and pragmatic solution, ensuring maximum safety while still seeking efficiencies.

On greener, SDG Sustainable Development Goals, this could be a key step towards achieving the industry's overarching People, Planet, and Profit (PPP) objectives, proving environmental responsibility can coincide with enhanced economic viability.

Industry Adoption and Challenges

The adoption of a waiver to hydrotest has been slow, as hydrocarbon leaks during operation are considered a significant risk. Having conventional, proven hydrotesting of the pipeline is considered a normally acceptable practice, with no risk of a learning curve. However, as the technology continues to improve and regulatory frameworks evolve, it is expected that more projects will transition to alternative testing methods. Table 6 presents large-diameter pipelines that have been installed without field hydrotesting (Pieter Swart, 2020).

Table 6—Pipeline Installed without Field Hydrotest

Year	Subsea Pipeline	Length (km)	Depth (m)	Pressure Design	Diameter (inch)
2003	Gulf of Aqaba	14.5	860	100	36
2004	Gulf of Terra	122	600	198	18
2019	Turk Stream	2 × 930	2200	300	32
> 2020	Nord Stream 2	2 × 1220	200	220	48

Conclusions

As the oil and gas industry navigates an era marked by sustainability initiatives, cost reduction, and digital innovation, traditional methods, such as subsea hydrotesting, must be re-examined considering emerging alternatives. This paper presents a comprehensive framework for waiving hydrotesting in subsea pipeline projects, one that is primarily developed based on risk-based ALARP principles, enabled by digital tools, and supported by empirical data.

Hydrotesting has served the industry well for decades, offering a straightforward method to verify hoop stress capacity and detect gross construction defects. However, its limitations, ranging from narrow failure mode coverage to environmental and logistical burdens, make it less suited for today's dynamic offshore projects. Moreover, field evidence and failure statistics confirm that many critical risks fall outside the scope of hydrotest verification.

In contrast, alternative assurance methods offer a predictive, comprehensive, and sustainable approach. Techniques such as mill testing, high-resolution NDT, baseline ILI, digital twin modelling, and QRA collectively form a robust verification matrix. When applied together, they not only replicate but often exceed the assurance level provided by hydrotesting while reducing commissioning timelines, environmental footprint, and operational costs. and elimination of chemical disposal, aligning with People, Planet, and Profit (PPP) objectives.

Key findings indicate that conventional subsea pipeline hydrotesting raises significant environmental concerns, including millions of gallons of water wastage, hundreds of tons of CO2 emissions, waste generation, contaminated water, and the discharge of disposable biocides and corrosion inhibitors, which disrupt sensitive habitats (e.g., coral reefs). UN Sustainable Development Goals (SDG)- 13 and SDG (14) recommend climate action, innovation, and responsible consumption of resources to achieve sustainable life below water.

Ultimately, the waiver of hydrotesting is not about circumventing safety it is about achieving safety through smarter, more efficient means. It reflects a broader industry transition from prescriptive testing toward performance-based, data-driven assurance. When combined with transparent documentation, strong quality controls, and cross-disciplinary collaboration, the hydrotest waiver becomes a strategic enabler, not a compromise.

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